

Held-out ecological prediction: Palmer penguins

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1 Overview

The earlier scoring examples used simulated cases to demonstrate machinery. Here the observations are real measurements of adult penguins collected around Palmer Station in 2007-2009 ([Gorman et al. 2014](#)). We fit a small finite Bayesian network on 2007-2008 and predict species in the untouched 2009 observations.

The model is a deliberately simple classifier built for this vignette, not a published model from the zoo. The question is predictive discrimination within this sampling setting, not a causal effect or a conservation recommendation.

2 Setup

The CC0 palmerpenguins table ([Horst et al. 2020](#)) is included verbatim beside this source in `data/penguins.csv`, with its source commit, SHA-256 and attribution in `data/LICENSE.md`. No model download or user's model cache is involved, so this vignette runs offline.

```

using CSV
using Statistics
using BayesianNetworks
using BayesianNetworkInference

rows = [NamedTuple(row) for row in CSV.File(joinpath(@_DIR_, "data",
"penguins.csv"));
                                                missingstring = "NA")]

complete = filter(rows) do r
    !any(ismissing, (r.species, r.bill_length_mm, r.flipper_length_mm, r.sex,
r.year))
end
training = filter(r -> r.year < 2009, complete)
held_out = filter(r -> r.year == 2009, complete)
(available = length(rows), complete_cases = length(complete),
excluded = length(rows) - length(complete), training = length(training),
held_out = length(held_out), training_years = sort(unique(r.year for r in
training)))

```

```

(available = 344, complete_cases = 333, excluded = 11, training = 216, held_out =
117, training_years = [2007, 2008])

```

2.1 Provenance and a fixed split

Rows missing a required predictor are excluded, with the count reported above. That is a complete-case analysis, not an assumption that missingness is harmless. All 2009 rows are kept together. No discretization cut point or CPT is learned from them.

3 Learn the representation using training data only

Bill and flipper lengths are discretized at training-set tertiles. Three bins and Laplace smoothing with pseudocount one are fixed choices for this demonstration; we do not tune them against the held-out scores.

```

bins = [:small, :medium, :large]
species = [:Adelie, :Chinstrap, :Gentoo]
sexes = [:female, :male]
bill_cuts = quantile([r.bill_length_mm for r in training], [1 / 3, 2 / 3])
flipper_cuts = quantile([r.flipper_length_mm for r in training], [1 / 3, 2 / 3])
bin(x, cuts) = bins[x <= cuts[1] ? 1 : x <= cuts[2] ? 2 : 3]
as_case(r) = Dict(:Species => Symbol(r.species),
                  :BillLength => bin(r.bill_length_mm, bill_cuts),
                  :FlipperLength => bin(r.flipper_length_mm, flipper_cuts),
                  :Sex => Symbol(r.sex))
train_cases = as_case.(training)
test_cases = as_case.(held_out)
(bill_cuts = bill_cuts, flipper_cuts = flipper_cuts)

```

```
(bill_cuts = [40.9, 46.43333333333333], flipper_cuts = [191.0, 209.0])
```

The classifier assumes the three predictors are conditionally independent given species. This naive-Bayes assumption is an approximation: the measurements are biologically related. It is not a claim about causal direction.

```
bn = bayesnet(:Species ⇒ species, :BillLength ⇒ bins, :FlipperLength ⇒ bins,
              :Sex ⇒ sexes;
              mechanisms = [:BillLength ⇒ :Species, :FlipperLength ⇒ :Species,
                           :Sex ⇒ :Species])
function fit_classifier(cases)
    prior_counts = [1.0 + count(c -> c[:Species] == s, cases) for s in species]
    fitted = bind_cpt(BayesModel(bn), :Species ⇒ prior_counts ./ sum(prior_counts))
    for (variable, levels) in ((:BillLength, bins), (:FlipperLength, bins), (:Sex,
sexes))
        counts = ones(length(species), length(levels))
        for c in cases
            i = findfirst(==(c[:Species]), species)
            j = findfirst(==(c[variable]), levels)
            counts[i, j] += 1
        end
        fitted = bind_cpt(fitted, variable ⇒ counts ./ sum(counts; dims = 2))
    end
    return fitted
end
model = fit_classifier(train_cases)
validate(model; closed = true, unique_names = true, semantics = true)
```

4 Score untouched observations against a prior baseline

predict withholds the target even though each case includes its observed label. The baseline ignores predictors and uses only the species prior learned from the same training cases. Lower Brier score, higher log score and higher accuracy are better.

```
prediction = predict(model, test_cases, :Species;
                    evidence_vars = [:BillLength, :FlipperLength, :Sex])
prior_prediction = baseline(model, test_cases, :Species)
results = [(method = label, n = length(p), brier = brier_score(p),
            log_score = log_score(p), accuracy = accuracy(p))
           for (label, p) in (("naive Bayes", prediction), ("training prior",
prior_prediction))]
results
```

```
2-element Vector{@NamedTuple{method::String, n::Int64, brier::Float64,
log_score::Float64, accuracy::Float64}}:
 (method = "naive Bayes", n = 117, brier = 0.1768949277634653, log_score =
-0.3298607488958938, accuracy = 0.8632478632478633)
 (method = "training prior", n = 117, brier = 0.6378120256896763, log_score =
-1.0530985668347588, accuracy = 0.4444444444444444)
```

```
confusion_matrix(prediction)
```

ConfusionMatrix over 3 states (accuracy 0.8632)

observed \ predicted	Adelie	Chinstrap	Gentoo
Adelie	43	8	1
Chinstrap	2	20	2
Gentoo	0	3	38

Aggregate accuracy can conceal differences between species, which is why the confusion matrix accompanies the proper scores. The same queries through a junction tree agree numerically with variable elimination:

```
via_tree = predict(model, test_cases, :Species;  
                  evidence_vars = [:BillLength, :FlipperLength, :Sex],  
                  backend = JunctionTree())  
(brier_difference = brier_score(prediction) - brier_score(via_tree),  
 log_score_difference = log_score(prediction) - log_score(via_tree))
```

```
(brier_difference = -1.6653345369377348e-16, log_score_difference =  
1.1102230246251565e-16)
```

Backend agreement establishes computational consistency on these cases, not predictive validity. The held-out labels provide the separate predictive evidence.

5 Summary: what this result does and does not establish

This is a temporal holdout with training-only preprocessing and an explicit baseline, rather than a fit scored on its own training observations. It remains one small dataset from one region. We have not established performance on new islands, different sampling protocols or future environmental conditions; the simplified table also does not let us audit repeated-individual dependence. The complete-case restriction, coarse bins and conditional-independence assumption are substantive limitations.

No uncertainty interval or population-level superiority claim follows from a single holdout score. A scientific deployment would need a prespecified target population, appropriate grouping, missing-data treatment, calibration assessment and external validation. The practical lesson is to refit every learned preprocessing step inside the training split and keep the evaluation question separate from whether inference returns the right number for the fitted model.

6 References

Gorman, Kristen B., Tony D. Williams, and William R. Fraser. 2014. “Ecological Sexual Dimorphism and Environmental Variability Within a Community of Antarctic Penguins (Genus *Pygoscelis*).” *PLOS ONE* 9 (3): e90081. <https://doi.org/10.1371/journal.pone.0090081>.

Horst, Allison Marie, Alison Presmanes Hill, and Kristen B. Gorman. 2020. *Palmerpenguins: Palmer Archipelago (Antarctica) Penguin Data*. <https://doi.org/10.5281/zenodo.3960218>.